

Subat Ahsan

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Education

University of Toronto

Sep 2022 – Apr 2027

Bachelor of Applied Science in Computer Engineering + PEY Co-op

- 3.42 / 4.0 GPA
- **Relevant Courses:** Data Structures and Algorithms, Programming Fundamentals, Computer Organization, Software Design and Communication, Computer Networks, Operating Systems

Skills

Languages: Python, C, C++, JavaScript TypeScript, HTML, CSS

Tools & Frameworks: AWS, Node.js, React.js, Express.js, Flask, FastAPI, MySQL, MongoDB, Git, Linux

Work Experience

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Software Engineer September 2025 – August 2026

- Developed complex web-based user interfaces using **React**, implementing and managing intricate component and application state.
- Integrated frontend components with backend APIs using **TanStack Query**, managing server state and ensuring data was efficiently fetched, cached, and synchronized within the application.
- Designed **DynamoDB** entities and access patterns to support efficient retrieval of data across new use cases, improving the flexibility and efficiency of the backend.
- Developed reusable **Python** wrappers around external APIs, including **Google Cloud** and Google Places API, simplifying their integration and use across backend services

Projects

Pokémon Showdown Teams | [GitHub](#)

- Developed a **full-stack application** for storing and sharing Pokémon Showdown teams
- Built a **RESTful API** with **Express.js** to handle authentication, team uploads, and data retrieval
- Developed a responsive **React.js** frontend with dynamic routing and seamless UI
- Implemented user creation and secure user authentication using **JWT** tokens
- Integrated **MongoDB** with Mongoose to efficiently store and retrieve user profiles and team data

City Mapping Software | Course: Software Design and Communication Feb – Apr 2024

- Collaborated with a team of 3 to develop mapping software using **C++**, **GTK** and OpenStreetMap's API
- Created graphical program that efficiently draws city maps using advanced rendering techniques
- Applied **object-oriented programming** to manage map data, rendering, and user interaction
- Implemented and fine-tuned algorithms such as Dijkstra's, A*, and Nearest Neighbor with Simulated Annealing to power efficient route-finding functionality
- Gained experience in team-based software development and codebase organization using Git

OS/161 Operating System Implementation | Course: Operating Systems Feb – Apr 2025

- Collaborated with teammate to implement core operating system components such as locks, system calls and semaphores using **C** and **Assembly** on System/161 machine simulator
- Debugged code by tracing kernel-level issues through low-level logs, stack traces, and memory analysis
- Greatly improved ability to comprehend, navigate and modify large and robust codebases